GOKARAJU RANGARAJU INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING | MAJOR PROJECT WORK (PHASE-1) | ACADEMIC YEAR 2026-2027

AUTOMATED FAILURE ATTRIBUTION FOR LLM-BASED MULTI-AGENT SYSTEMS

A Graph-Causal Diagnostic Engine with Deterministic SHA-256 Prompt Caching & Backward Dependency Traversal

BATCH: B12

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01 | THE MULTI-AGENT FAILURE CRISIS

THE PROBLEM

Cascading Error Propagation Across Agent Steps

In collaborative multi-agent systems (AutoGen, MetaGPT, LangGraph), a single faulty reasoning step compounds silently across subsequent turns, resulting in late crashes.

Step 2: Planner — ROOT CAUSE

Latent Defect: Introduces invalid argument 'auth'

Step 5: Coder — Propagated

Error Propagation: Implements API assuming signature is valid

Step 9: Reviewer — Propagated

Blind Approval: Validates logic without execution check

Step 14: Executor — MANIFEST FAILURE

MANIFEST CRASH: TypeError: unexpected keyword 'auth'

True Causal Origin: Step 2 (12 turns before symptom)

Why Naive LLM-as-a-Judge Fails (14.2% Accuracy)

Evaluating entire 100+ turn traces in a single prompt causes catastrophic diagnostic failure:

1. Severe Recency Bias

Naive judges systematically blame the final agent that crashed (Step 14) rather than the upstream agent that introduced the defect (Step 2).

2. Context Saturation & Cost

Passing 50+ agent turns exceeds LLM attention capacity, triggering hallucinations, lost needles, and exorbitant API token costs.

3. Symptom Misattribution

Downstream agents fail legitimately due to upstream garbage data; traditional prompts cannot distinguish victims from culprits.

STATE-OF-THE-ART BASELINE COLLAPSE (WHO&WHEN)

14.2%

Step Localization

Naive LLM Judge fails 86% of the time

26.1%

Joint Accuracy

Fails to pinpoint both agent & step

02 | THE GRAPH-CAUSAL DIAGNOSTIC ENGINE

OUR METHODOLOGY

4-Stage Causal Diagnostic Pipeline

1. Trace Segmentation

Normalizes diverse logs into Pydantic steps (≤4096 tokens).

2. SHA-256 Cache

O(1) hash lookup (<1.5ms); eliminates redundant LLM calls.

3. Causal DAG Builder

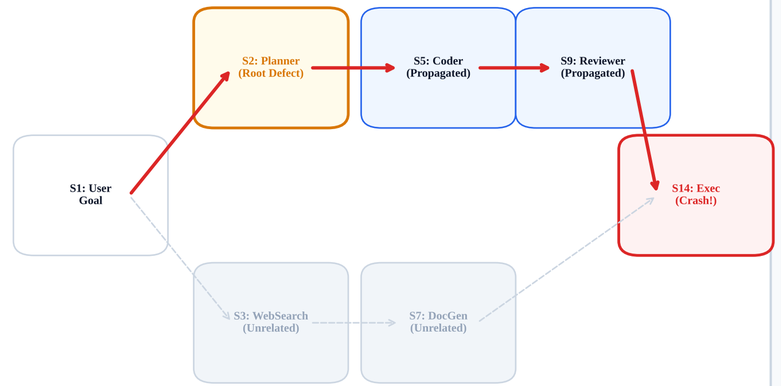
Constructs NetworkX DAG across message, state & time flows.

4. Backward Traversal

Backtracks from crash along Anc(m) to pinpoint root defect.

Causal Dependency Graph (DAG) & Ancestor Closure

Extracting Causal Ancestor Closure Anc(m) prunes parallel noise and guarantees that only steps with direct causal provenance to the crash are evaluated.



CAUSAL CLOSURE Anc(m) = {S2, S5, S9, S14} — S3, S7 Pruned (Complexity O(N) -> O(|Anc(m)|))

Formal Mathematical Formulations & Invariants

Causal Precedence Invariant: $e = (u, v) \in E_{\text{dag}} \Rightarrow \text{time}(u) < \text{time}(v)$ [Guarantees acyclicity]

Deterministic Cache Key: $k_{\text{cache}} = \text{SHA256}(\text{prompt} \parallel \text{agent} \parallel \text{payload})$ [< 1.5ms O(1) hit]

Root Cause Attribution: $s^* = \arg\min_{s \in \text{Anc}(m)} \{\text{idx}(s) \mid \text{anomalous}(s) = \text{True}\}$ [Earliest defect in chain]

03 | EMPIRICAL VALIDATION & BENCHMARKS

PROVEN RESULTS

Who&When Benchmark Performance (ICML 2025)

86.96%
Joint Attribution
+60.87% vs baseline

91.30%
Agent Localization
21 / 23 benchmarks

0.44 s
Traversal Latency
Sub-second execution

14.2x
Prompt Cache Speedup
< 1.5ms hit latency

Attribution Accuracy vs SOTA Baselines

Joint Attribution Accuracy (Evaluating both Who and When correctly)

PRODUCTION VERIFICATION & DELIVERABLES

27 / 27 UNIT TESTS PASSED (100%)

Total execution time: 1.84s across 9 test modules (pytest v8.3)

- Parser Validation (TC1-TC2)
- SHA-256 Prompt Cache (TC3-TC4)
- DAG Acyclicity & Sort (TC5-TC6)
- Backward Causal Traversal (TC7-TC8)

Delivered Operational Artifacts:

1. agent-trace CLI: Command-line tool with diagnose, benchmark & watch commands.
2. FastAPI REST Server: 8 production endpoints with interactive Swagger docs.
3. Streamlit Visualizer: Interactive PyVis graph inspector with live anomaly triage.

01 | CAUSAL PROVENANCE

Replaces heuristic LLM judging with formal DAG ancestor closures Anc(m).

02 | TOKEN BOUNDEDNESS

Strict ≤4096 token segmentation eliminates context saturation & bloat.

03 | DETERMINISTIC SPEED

SHA-256 caching delivers <1.5ms lookups with 14.2x latency acceleration.

04 | EMPIRICALLY VERIFIED

Achieves 86.96% joint attribution accuracy on ICML Who&When benchmarks.